



CAMV: A Framework for Context-Aware Multi-View Visualization of Data Analysis Results

Yanna Lin¹ · Liwenhan Xie¹ · Leixian Shen¹ · Zhuochen Jin² · Sicheng Song⁵ · Zikun Deng³ · Huamin Qu¹ · Ke Xu⁴

Received: 5 October 2025 / Revised: 1 March 2026 / Accepted: 6 March 2026
© The Author(s) 2026

Abstract

Understanding analytical results in visual data analysis is essential to inform further exploration and decision-making. However, existing tools often visualize the results alone, offering limited access to associated contexts needed to explain why the results occur. To address this gap, we propose CAMV, a framework that automatically extracts relevant contextual information from given results and datasets, and generates multi-view visualizations to support deeper understanding. CAMV consists of two major components: one that identifies key data subspaces and fact types as explanatory context, and the other that generates coordinated multi-view visualizations of both the analysis results and their contextual information. Based on the framework, we developed an interactive prototype and conducted a comparison study with 12 participants, using GPT-4o as a baseline. Results show that CAMV enhances understanding, decision-making speed, and trust compared to GPT-4o, with positive feedback on its data-grounded reasoning and structured output.

Keywords Context-aware multi-view visualizations · Dashboard · Visual explanation of analytical results · User intent inference

✉ Ke Xu
xuke@nju.edu.cn
Yanna Lin
ylindg@connect.ust.hk
Liwenhan Xie
liwenhan.xie@connect.ust.hk
Leixian Shen
lshenaj@connect.ust.hk
Zhuochen Jin
zcjin.idvx@gmail.com
Sicheng Song
scsong@dase.ecnu.edu.cn
Zikun Deng
zkdeng@scut.edu.cn
Huamin Qu
huamin@cse.ust.hk

- ¹ Hong Kong University of Science and Technology, Hong Kong, China
- ² Tongji University, Zhejiang, China
- ³ South China University of Technology, Guangdong, China
- ⁴ Nanjing University, Jiangsu, China
- ⁵ East China Normal University, Shanghai, China

1 Introduction

Understanding analytical results, such as outliers and clusters, is critical for guiding data cleaning, exploration, and decision-making [1]. Interpreting why these results occur often requires additional context, such as underlying feature distributions [2], correlations [3, 4], and specific data subspaces (i.e., data subsets filtered by specific attribute combinations) [5]. However, users often lack access to this necessary context, as analytical results (e.g., cluster labels or outlier flags) are typically detached from the data attributes or patterns that are necessary for interpretation. While existing eXplainable AI (XAI) methods like SHAP [6] facilitate interpretation by quantifying feature importance scores, recent work has demonstrated that attribution alone is insufficient [2]. Result interpretation requires surfacing contextual evidence, such as the actual data distributions and subspace patterns, that allows users to verify and reason about why a result occurred. Although systems like VBridge [2] attempt to provide such context, the context is manually defined and domain-specific. Consequently, users are left to explore potentially helpful data subspaces by trial and error, which is tedious and time-consuming, especially in high-dimensional datasets.

Multi-view visualizations (MVs) offer a promising solution for result interpretation by presenting multiple coordinated visualizations on a single screen, allowing users to examine diverse data perspectives simultaneously [7–9]. Unlike traditional visualization tools (e.g., Vega-Lite [10] and Matplotlib [11]) that visualize results in isolation, MVs can reveal not only the results themselves but also the contextual evidence necessary for understanding them. For example, to understand why a cluster emerges, MVs can reveal feature distributions, contributions, and metric differences across groups, enabling users to explore and verify the reasons behind the clustering results.

However, constructing such MVs remains challenging. Existing MV creation approaches, both data-driven and intent-driven, often fall short in supporting result interpretation because they are primarily designed for general data exploration rather than targeted result interpretation. Data-driven methods, such as MultiVision [12] and DashBot [13], typically recommend optimized sets of visualizations based on empirical design guidelines, e.g., using multiple views minimally and diversely. While effective for finding unknown patterns, these methods are result-agnostic. They often surface globally interesting views that lack a direct logical connection to the specific analytical result a user aims to explain. To address this limitation, intent-driven methods like Medley [14] and TaskVis [15] allow users to guide MV recommendations by explicitly selecting analysis tasks and data columns of interest. However, these methods assume that users have prior knowledge of which data attributes are relevant. This poses significant challenges in real-world applications, particularly when users are trying to interpret unfamiliar or unexpected results, or when they are uncertain about which tasks or data would aid their understanding. With the advancement of large language models (LLMs), recent research has explored leveraging LLMs to enhance visual analytics (a type of MVs) but often not for focused result interpretation [16–18]. These systems remain primarily process-oriented, focusing on facilitating analytical workflows (e.g., proactively providing assistance [18] or orchestrating task-driven exploration [17]) or lowering interaction barriers (e.g., visual onboarding [16]). As a result, they often lack a mechanism to automatically infer the context required to explain a specific analytical result.

To address the challenges, we propose Context-Aware Multi-View Visualization (CAMV), which treats analytical results as user intent to automatically infer relevant data subspaces that provide context for understanding the results. CAMV then recommends MVs to visualize these subspaces. Unlike previous systems, CAMV eliminates the need for users to manually specify analysis tasks or attributes, instead using the analytical results themselves, along with original data and metadata (such as applied algorithms),

as cues to determine the most relevant context for visual understanding. Focusing on analysis tasks with discrete result outputs (e.g., clustering, outlier detection, classification), CAMV stands apart from prior systems by automating the entire process of context generation and visualizing such contexts as coordinated MV. CAMV comprises two modules: the *analytical intent inference module* and the *multi-view visualization module*. The inference module identifies subspaces that help understand the results by first augmenting the result table with contextual attributes derived from the original dataset. Each feature is then scored based on its impact on the result, its correlation with key metrics (i.e., the attributes used for deriving analytical results), and its distributional divergence across result groups. From the top-ranked features, CAMV constructs subspaces with corresponding statistical data facts (i.e., specific characteristics of data, such as an increasing trend or a normal distribution), forming candidate intents for visual explanation. The visualization module then recommends appropriate multi-view visualizations for these subspaces, using rule-based mappings informed by data semantics and analytical context. We implement a prototype interface of CAMV and validate its usability and effectiveness through system performance analysis, a use case walkthrough, a user study with 12 participants compared with ChatGPT, and a qualitative comparison. In summary, our contributions are as follows:

- We introduce CAMV, a novel result-oriented framework that automatically infers the context required for result interpretation by transforming the user’s high-level interpretation intent into low-level relevant data subspaces, and then visualizes these subspaces as coordinated MVs.
- We design and implement a system that enables users to specify results and metadata, and explore recommended visualizations through an interactive interface.
- We evaluate the usability and effectiveness of CAMV via a use case, a comparison user study, system performance measures, and a qualitative comparison.

2 Related Work

2.1 Multi-View Visualization Creation

Existing techniques for creating MVs can be divided into two groups, namely data-driven and user intent-driven.

Data-driven approaches usually follow empirical research guidelines [7, 19–23], recommending multi-view visualizations by optimizing dashboard design goals [12, 13, 24–27]. For example, MultiVision [12] is a hybrid human-computer system. It integrates state-of-the-art multi-view

visualization guidelines to optimize the generation of visualizations, overcoming the limitations of traditional single-view recommendation methods. Additionally, Dashbot [13] employs deep reinforcement learning to construct training environments and reward mechanisms based on existing theories, effectively simulating human exploration behavior to automatically generate efficient analytical dashboards. Li et al. explores user sentiments and topic distributions in P2P accommodation reviews [28, 29]. While these systems generate optimal visualizations aligned with their objectives, they often overlook specific user intents, potentially failing to meet actual user needs.

User intent-driven approaches usually allow users to express interest based on their data analysis tasks and then the system automatically generates corresponding MVs [14, 15, 30, 31]. For example, Medley [14] enables users to select data columns and specific analysis tasks, recommending a set of visualizations for each task. Medley features a lightweight interface for users to choose from recommended visualizations and configure interactive dashboards on a separate panel. Similarly, TaskVis [15] offers a task-oriented visualization recommendation system, allowing users to manually select analysis tasks and data columns of interest. It uses rule-based ranking to generate multi-view visualizations aligned with user intentions. While these systems account for user intents, they require users' prior knowledge of the analysis tasks and data, and need additional interactions to express intents.

Recently, large language models (LLMs) have been increasingly integrated into visual analytics (one type of MVs) to provide intelligent assistance throughout the analysis workflow. For instance, *LEVA* [16] enhances the VA workflow across multiple stages, including onboarding, exploration, and summarization, by interpreting system specifications and recommending insights. *ProactiveVA* [18] introduces an LLM-powered UI agent that proactively assists analysts by detecting help-seeking behaviors and offering contextual guidance. Similarly, *LightVA* [17] employs an agent-based recursive workflow (consisting of a planner, executor, and controller) to reduce the complexity of task-driven data exploration. However, these LLM-based frameworks are primarily *process-oriented*, focusing on facilitating the iterative analysis pipeline or assisting in task planning and system navigation. They are not specifically designed for *result interpretation*; that is, they do not focus on identifying the explanatory data contexts or subspaces that underlie a specific, already-derived analytical outcome.

In summary, while existing multi-view visualization methods excel at facilitating data exploration or MV understanding and navigation, they largely operate in a result-agnostic manner. In practice, users often derive preliminary results through exploration and seek further exploration

to interpret these results. Thus, we propose an innovative result-oriented framework in this paper, utilizing users' data analysis outcomes as the basis for expressing intent, to automatically infer and recommend MVs that provide the necessary explanatory context for interpreting specific results.

2.2 Analysis Intent Inferring

User intent plays a crucial role in data analysis, as a system that effectively understands user intent can avoid generating one-size-fits-all results. Understanding user intent involves two key aspects: how users provide intent information and how systems interpret it.

Existing techniques have introduced various paradigms for users to express their analytical needs, including natural language [32], sketches [33, 34], WIMP [15, 35], examples or references [36, 37], and behaviors [38, 39]. For instance, WIMP serves as a fundamental interaction paradigm, allowing users to manipulate icons or menu options to select and arrange visual elements, adjust parameters, or specify desired representations [35]. Similarly, users can sketch or annotate directly on visual elements or canvases to indicate desired modifications, annotations, or interactions. These sketches are then interpreted and transformed into meaningful actions or visual enhancements [33]. Natural language-based paradigms have become increasingly popular with the rise of large language models, enabling users to express their needs freely and flexibly in natural language [32]. The system extracts user intent from the queries and performs corresponding tasks. Intent inference is also increasingly critical in visual trust and security, such as deducing the underlying motives behind misleading visualizations [40] and tampered visualizations [41]. While these paradigms effectively support intent expression, they often require manual effort from users to specify their intent explicitly.

Another category of approaches seeks to infer user intent automatically from user behaviors [38, 39]. These methods observe users' interactions, such as navigation, selection, or modification of visual elements, to uncover implicit intents. For example, Solas [38] tracks users' interaction history during data analysis and provides task-specific visualization recommendations. Lalle et al. [39] proposes a gaze-driven adaptive mechanism to assist users in comprehending magazine-style narrative visualizations. While behavior-based paradigms reduce the need for explicit user input, they rely heavily on extensive interaction history.

To address these limitations, this paper proposes a novel intent expression paradigm that uses users' analysis results as intent information. The system then interprets these results to automatically generate corresponding outputs, reducing the need for extensive user interaction or historical data.

2.3 Explainable AI Methods

Interpretability is a cornerstone of trustworthy visual analytics. Research in eXplainable AI (XAI) is generally categorized into *intrinsic interpretability* and *post-hoc explanations* [2]. Intrinsic methods rely on inherently transparent models, such as decision trees or linear regression, where the model structure directly reveals the logic behind a prediction [42, 43]. However, as analytical tasks and corresponding models grow in complexity, researchers increasingly turn to post-hoc methods to interpret results of opaque black-box models or high-dimensional data mining processes.

Post-hoc explanations can be further divided into model-level and instance-level approaches. Model-level methods, such as RuleMatrix [44] or Mimic-learning [45], use interpretable machine learning models such as gradient boosting trees or decision trees to mimic the complex models and provide rule explanations. However, they focus on the model explanations but often neglect the nuances of specific analytical outcomes. Instance-level methods, such as SHAP [6] and LIME [46], focus on *feature attribution* to quantify the contribution of each input feature to a specific result. While these methods are effective at identifying *what* features are important, they often fail to explain *why* these features lead to a particular outcome. Users frequently find it challenging to connect abstract importance scores to the underlying data distribution or real-world context. To bridge this gap, systems like VBridge [2] attempt to link machine learning features back to the original data to assist domain experts. However, such tools often rely heavily on experts to manually define relevant contexts and identify meaningful associations, which limits their scalability and automation in general data analysis scenarios.

Different from traditional XAI methods that prioritize model structure or simple feature attribution, CAMV focuses on result interpretation by automatically generating explanatory context. Instead of requiring manual intervention from domain experts, our framework automatically identifies key data subspaces that serve as the most

relevant background for a given result. Furthermore, while existing XAI techniques often provide static or limited visualizations, CAMV dynamically generates coordinated multi-view visualizations tailored to the specific analytical outcome. This data-grounded reasoning approach ensures that the explanations are not only automated but also deeply rooted in the dataset's context.

3 Methodology

We propose a new framework, Context-Aware Multi-View Visualization (CAMV), as illustrated in Fig. 1. It utilizes the data table and its associated analysis results as input. Through an automated inference process, it discerns the user's subsequent exploratory intentions for understanding results (Sect. 3.1). Then, it generates MVs that provide meaningful contextual information, facilitating a more comprehensive understanding of the findings in the analysis results (Sect. 3.2). Please refer to the supplementary material for the pseudo code and computation complexity analysis.

3.1 Analytical Intent Inference

As illustrated in Fig. 1, CAMV infers the user's analytical intent based on both analytical results and the original data context. Specifically, the analysis intent is defined as the combination of multiple data subspaces, each associated with its corresponding data facts. For example, Fig. 4(a) shows one such data subspace, [*avg_Number_of_Male2*, *Number_of_Dead2*, *cluster*], along with its corresponding data fact, *Correlation*. CAMV automatically identifies the analytical intent through two main components: (1) a Feature Enhancement Module, enriching analytical result tables with contextual feature columns extracted from the original data tables to build up an augmented analysis table for subsequent analytical exploration; (2) an Analysis Intent Construction Module, which generates meaningful analysis intent according to the augmented table; To

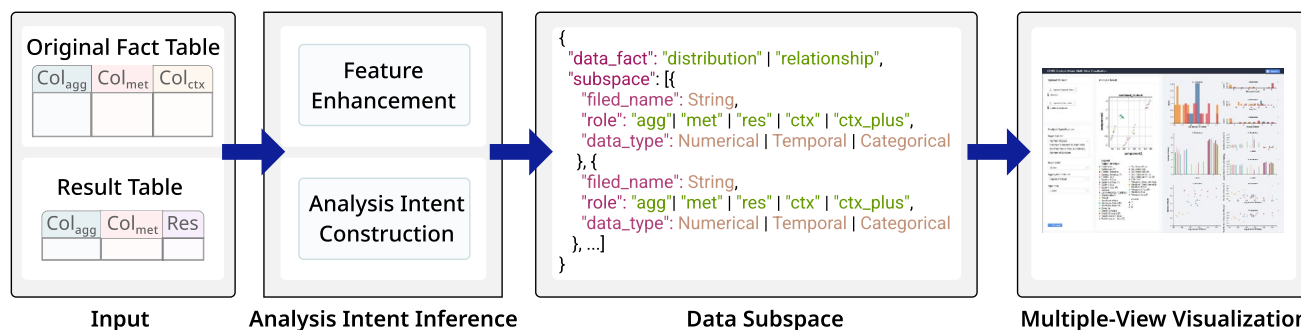


Fig. 1 The pipeline of CAMV. Given the analysis results with fact table as input, CAMV infers the analytical intent as data subspaces and corresponding fact types, and then visualizes these subspaces through coordinated multi-view visualizations

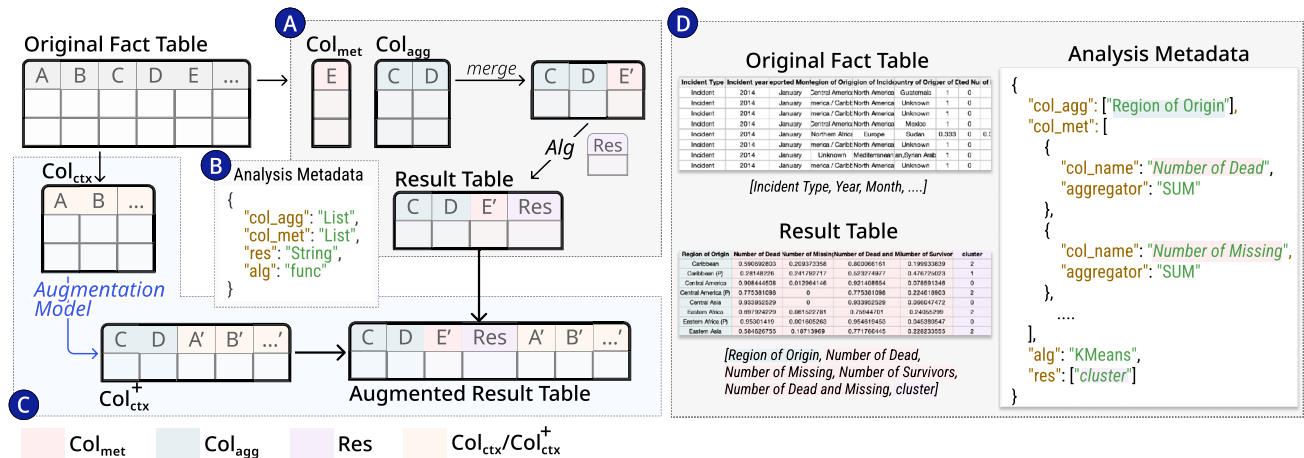


Fig. 2 The figure illustrates: (A) the analysis pipeline that generates the result table from the original fact table; (B) the corresponding analysis metadata; (C) the feature enhancement module, which augments the result table by augmenting contextual attributes derived from the

elaborate further, this module can be divided into two sequential steps. The initial step evaluates significance of each features and selects the top- M most relevant ones from the augmented table. Subsequently, the second step generates analysis intents with potential subspaces and specific data facts based on those important features, evaluates the interestingness of intents, and selects the top- N intents with most meaningful information for interpreting the analysis results.

In the following, we first formalize the input data representation (Sect. 3.1.1), followed by detailed descriptions of the two key modules, Feature Enhancement Module in Sect. 3.1.2 and Analysis Intent Construction Module in Sect. 3.1.3.

3.1.1 Input Formulation

To support consistent processing across diverse data analysis tasks, we formalize the input as a structured JSON object that captures the essential elements of typical analysis workflows (shown in Fig. 2(A)), as analysts often apply various operations to derive analytical results from original datasets.

Through collaboration with 7 analysis experts from a leading company analysis department, we identified a common analysis pipeline for generating analytical results from raw data. The process begins with an original fact table that records raw business events. Analysts first select metrics, which represent key analytical indicators of interest—such as *profit*, *sales*, or *cost*—from the fact data. Next, they specify a set of aggregation attributes (col_{agg}), such as region or time period, to determine the level of analysis for these metrics. Aggregation operations (e.g., *sum*, *avg*) are then applied to summarize the selected metrics at the specified levels, producing aggregated metric values (col_{met}) for subsequent

original fact table and then adding it; and (D) an example showing the original fact table, result table, and corresponding analysis metadata, helping to explain the concepts of col_{agg} , col_{met} , and res

Table 1 Notations in analytical intent inference module

Notation	Description
col_{agg}	Aggregation attributes that define the analysis level (e.g., region, time)
col_{met}	Metric attributes representing key indicators of interest (e.g., sales, profit)
res	Algorithm output column in the result table (e.g., outlier label)
col_{ctx}	Context attributes from the original fact table that are not present in the result table
col_{ctx}^+	Augmented contextual features derived from col_{ctx} using statistical or distributional operations
alg	Analytical algorithm applied to generate res

analysis. After that, specific analytical algorithms (denoted as alg) are applied to these aggregated metrics, yielding the final analytical results (res). Thus, a result table is derived using alg and typically includes three types of columns, i.e., col_{agg} for aggregation levels, col_{met} for metrics of interest, and res for algorithm outputs. We organize this information into a structured JSON object comprising two components: the analysis metadata and the context metadata, as shown in Fig. 2(B). The remaining attributes not directly involved in the analysis, i.e., in the original fact table but not in the result table, are designated as context attributes col_{ctx} . The definitions of these notation are summarized in Table 1.

Figure 2(D) shows an example. The original fact table is the *Missing Migrants* dataset, with data attributes such as *Incident Type*, *Year*, and *Month*. The analyst aggregated the data by *Region of Origin* and performed clustering analysis based on key attributes, including *Number of Dead*, *Number of Missing*, *Number of Dead and Missing*, and *Number of Survivors*. The analysis categorized regions into three clusters labeled 0, 1, and 2, stored in *cluster*. In this example, the analysis metadata includes the aggregation attribute (col_{agg}

), which is *Region of Origin*; the metrics of interest (col_{met}), which are *Number of Dead*, *Number of Missing*, *Number of Dead and Missing*, and *Number of Survivors*; the algorithm output (res), which is *cluster*; and the analytical algorithm used, which is *KMeans*. Moreover, attributes not directly involved in the analysis, such as *Incident Type*, *Year*, and *Month*, are considered context attributes (col_{ctx}).

3.1.2 Feature Enhancement Module

In data analysis processes, result tables often contain limited information, which may lack sufficient context for interpreting the analytical results. To address this limitation, we propose an *augmentation model* that enhances the result table by incorporating contextual attributes col_{ctx} from the original fact table. This process generates an augmented result table containing derived contextual features (col_{ctx}^+) that support interpretation and follow-up exploration, as shown in Fig. 2(C).

To generate these features, the augmentation model applies different aggregation strategies based on the data type for each contextual attribute. All contextual attributes are aggregated along the same analysis level (i.e., col_{agg}) as the original analysis to ensure consistency and relevance. For categorical and temporal data, we compute the number of distinct values (COUNT_DISTINCT), and distributions, including absolute counts (COUNT_DISTRIBUTION) and percentages (PERCENT_DISTRIBUTION). For numerical data, we perform basic statistical aggregations such as avg, sum, min, max, and histogram-based distributions (COUNT_HISTOGRAM and PERCENT_HISTOGRAM). Each resulting feature is named to reflect the applied operation, such as avg_Deaths.

3.1.3 Analysis Intent Construction Module

Existing work does not provide a formal definition of what constitutes contextual information to understand analytical results. We conceptualize contextual information as the combination of data subspaces and data facts which indicates the potential analysis intent to explore the explanatory cues for why particular results occur. This conceptualization is grounded in empirical observations from both academic and real-world data analysis practices.

Table 2 Metrics for data attribute importance evaluation

Dimension	Evaluation metric	Score function
Feature impact on analysis results	Sensitivity to results	$s_{sensitivity}$
	Distribution divergence across results	$s_{divergence}$
Feature-metric relationship	Correlation with target metrics	$s_{correlation}$

We review case studies in academic papers about outlier analysis [3, 4, 47], examine real-world data analysis workflows from Kaggle notebooks about data mining to investigate how analysts conduct follow-up analysis after identifying outliers, and investigate how existing research support the following visual data analysis [31, 48, 49]. Across these sources, we observe common strategies, including (i) assessing feature contributions to model outputs, (ii) comparing attribute distributions across outcome groups (e.g., outliers vs. inliers, different clusters), and (iii) exploring the correlation of features with the key indicators.

In implementing these strategies, we note a practical challenge: real-world datasets typically contain many features, and it is impractical to apply all strategies exhaustively to each feature. To address this, we propose a two-step approach: (1) identifying a small set of features that are most likely to contribute to interpreting the analysis results through **feature importance evaluation**, (2) applying the three observed strategies to generate data subspaces with corresponding data facts (e.g., correlation, distribution divergence) for each important feature through **intent construction**. In the following sections, we will introduce how to automatically complete the above two steps to identify the most insightful analysis intents.

Feature Importance Evaluation. To evaluate the importance of each feature in the analysis process, we define a set of evaluation metrics based on three key perspectives, which directly correspond to the **observed strategies** for understanding the influence of features. These perspectives are: (i) Sensitivity Score: Measures how a feature impacts the analytical result by assessing how changes in the feature affect the outcome. (ii) Divergence Score: Evaluates how a feature's distribution varies across different result categories, quantifying its ability to distinguish between these categories. (iii) Correlation Score: Quantifies the statistical relationship between the feature and the metrics used in the analysis. Intuitively, a feature receives a higher overall score if it can (1) independently induce a similar outcome, (2) separate different result groups in its distribution, or (3) correlate with the key metrics. We then combine these three scores into a single importance score to rank features for downstream intent construction.

Based on this definition, we propose a feature importance evaluation model that quantifies each feature along these three key dimensions, as summarized in Table 2. The details of the evaluation models are shown as below.

- Sensitivity Score measures the influence of the feature on the analytical result:

$$s_{sensitivity} = 1 - Dis(res', res), \quad (1)$$

$$res' = alg(col_{ctx}^+), \quad (2)$$

where res represents the original analysis result, res' is the result generated by applying the analytical algorithm $alg(\cdot)$ to the candidate feature col_{ctx}^+ , and $Dis(\cdot)$ denotes the dissimilarity between them normalized to the range $[0, 1]$. Specifically, we compute res' by projecting the candidate feature col_{ctx}^+ into the original metric space, substituting one of the original metrics while neutralizing others via zero-masking. This ensures the algorithm receives the required input dimensionality while generating a result driven solely by the variance of col_{ctx}^+ , enabling a direct comparison with res . The dissimilarity function $Dis(\cdot)$ is computed using the $1 - \text{Adjusted Rand Index (ARI)}$ [50] or *Jaccard Distance* [51]. The choice of the dissimilarity function $Dis(\cdot)$ is adaptively determined by the analytical tasks. For partition-based tasks (e.g., clustering), where the goal is to assess the stability of data groupings regardless of label assignments, we employ $1 - \text{ARI}$ to measure the consistency of sample-wise associations. For subset-based outputs (e.g., outlier detection), where the focus is on the precise identification of specific target instances, we utilize *Jaccard Distance* to quantify the overlap between the identified subsets. Consequently, a smaller $Dis(\cdot)$ indicates a higher structural similarity between res' and res , signifying a greater independent influence of col_{ctx}^+ on the original analytical result.

- **Divergence Score** evaluates the degree of difference in the feature's distribution across various result categories

$$s_{divergence} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n JSD(Dist_{col}(res_i), Dist_{col}(res_j)), \quad (3)$$

where n is the number of clusters in the analysis result, $JSD(\cdot)$ denotes the Jensen–Shannon divergence between feature distributions, and the $Dist_{col}(res_i)$ represents the distribution of one specific feature in the res_i .

- **Correlation Score** quantifies the statistical association between the feature and each target metric:

$$s_{correlation} = \frac{1}{n} \sum_{i=1}^n Corr(col_{ctx}, col_{met_i}), \quad (4)$$

where n is the number of target metrics and $Corr(\cdot)$ is a correlation measure (e.g., Pearson or Spearman). The importance of each augmented contextual feature (col_{ctx}^+) is computed as a linear combination of three scores:

$$s_{col_{ctx}^+} = w_1 \cdot s_{sensitivity} + w_2 \cdot s_{divergence} + w_3 \cdot s_{correlation}. \quad (5)$$

In our implementation, we adopt an equal-weighted fusion strategy by setting $w_1 = w_2 = w_3 = 1$. This design choice is informed by analysis experts from a leading industry partner, who regard these three contextual perspectives as equally important signals when identifying informative contexts in practice. Based on the scores, we select the top- M most important features for downstream intent construction, where M is determined by user requirements and computational constraints.

Intent Construction. After identifying the top- M most important augmented contextual features (col_{ctx}^+), we generate explanatory intents by constructing corresponding data subspaces and associating them with appropriate data facts. These intents operationalize the three observed analysis strategies introduced earlier. Specifically, we define two types of explanatory analysis intents:

- **Impact of feature distributions** on analysis results, derived from strategies (i) and (ii). This focus on understanding how distributions of individual features vary across different result groups, operationalized as *distribution divergence*. For example, Fig. 4(a) illustrates how *avg_Number of Male* is distributed across clusters; users can easily discern that regions in Cluster 1 (yellow) exhibit a lower male rate, whereas regions in Cluster 0 (red) show a higher rate; and
- **Impact of feature–metric relationship** on analysis results, derived from strategy (iii). This involves understanding how the correlation between a feature and a metric, and how their joint distribution differs across result groups. Accordingly, the corresponding data facts include both correlation and joint distribution divergence. For example, Fig. 4(f) illustrates the relationship between *avg_Number of Male* and *Number of Dead*. Even if a strong general correlation exists, a high divergence in their joint distribution reveals that clusters occupy distinct functional areas: Cluster 1 (yellow) is concentrated in the lower-left, while Cluster 0 (red) shifts to the upper area. This indicates that the specific impact of a feature on the metric varies by group.

To instantiate these intents, we construct data subspaces by combining each selected augmented context feature with relevant columns such as the analysis result (res), metric columns (col_{met}), and aggregation attributes (col_{agg}), as summarized in Table 3. Each subspace is linked to a corresponding data fact type—either *distribution divergence* or *relationship divergence*—depending on whether it concerns

Table 3 Mapping from analysis intents to data subspaces and data facts

Analysis intent	Data subspace	Data fact
Impact of Feature Distribution on Results	$(col_{ctx}^+, res), (col_{ctx}^+, col_{agg}, res)$	Distribution divergence
Impact of Feature-metric Relationship on Results	$(col_{ctx}^+, col_{met}, res)$	Correlation & distribution divergence

feature-level or feature–metric variation across result groups.

We quantify the interestingness of each subspace using the **silhouette score** [52].

The choice of silhouette score as a proxy for explanatory strength is motivated by its ability to simultaneously evaluate inter-group separation and intra-group cohesion. Unlike metrics that only consider mean discrepancies (such as Euclidean distance [53]), the silhouette score ensures that a data subspace not only exhibits a distinct contrast between result groups (inter-group separation) but also maintains high internal consistency within each group (intra-group cohesion). While correlation is a data fact for *feature-metric relationship* intent, we primarily utilize the silhouette score to rank subspaces for two reasons. First, the initial selection of top- M features already ensures a strong overall correlation between the features and the metric, making further correlation-based score redundant. Second, the *feature distribution* intent is associated with only one data type, distribution divergence. Including correlation in the interestingness score would introduce bias towards the data subspaces of *feature-metric relationship* intent. A higher

silhouette score indicates stronger structural contrast, making the subspace more suitable for visual explanation.

Finally, we select the top- N most informative intents based on their interestingness scores for downstream multi-view visualization recommendation. Each intent is represented in a structured JSON format for processing, as illustrated in Fig. 1.

3.2 Multi-view Visualization

Building on established visualization design theories [54–56], we define a task-oriented multi-view visualization module that maps analytical subspaces to appropriate visual representations. This module selects suitable chart types and visual encodings based on three key characteristics of each data subspace: data fact type (e.g., distribution divergence, correlation), data type (e.g., numerical, categorical), and data role (e.g., aggregation column, metric) of the attributes. To operationalize these mappings, we encode the visualization design knowledges into a set of rule tables tailored to our analytical context, as shown in Table 4a and b. They present the visualization rules applied to scalar and vector data attributes, respectively. These tables guide how different subspace types should be visually represented.

Based on these rules, we define the visualization recommendation function as follows:

$$\text{vis_rec}(\text{data_fact}, \text{data_type}, \text{data_role}) \rightarrow (\text{chart_type}, \text{encoding}, \text{transform}). \quad (6)$$

Table 4 Visualization rules for scalar (top) and vector (bottom) data attributes

Data subspace	User intent	Data type	Chart type	X-axis	Y-axis	Color
(a) Visualization rules for scalar data attributes						
$[col_{ctx}^+, res]$	Distribution	N*C	Stacked Bar Chart	$\text{bin}(col_{ctx}^+)$	$\text{count}(col_{ctx}^+)$	res
	Divergence	T*C	Stacked Bar Chart	col_{ctx}^+	$\text{count}(col_{ctx}^+)$	res
$[col_{ctx}^+, col_{agg}, res]$	Distribution Divergence	N/T*C*C	Bar Chart	col_{agg}	col_{ctx}^+	res
$[col_{ctx}^+, col_{met}, res]$	Correlation	N*C*N	Multicolor Scatter Plot	col_{ctx}^+	col_{met}	res
		T*C*N	Parallel Coordinate Plot	$col_{ctx}^+ + col_{met} + res$		res
(b) Visualization rules for vector data attributes						
$[col_{ctx}^+, res]$	Distribution	$[N/T] * C$	Stacked Bar Chart	$\text{Dim}(col_{ctx}^+)$	$\text{Value}(col_{ctx}^+)$	res
	Divergence					
$[col_{ctx}^+, col_{agg}, res]$	Distribution Divergence	$[N/T] * C * N$	Stacked Bar Chart	$\text{Dim}(col_{ctx}^+)$	$\text{Value}(col_{ctx}^+)$	res
			Parallel Coordinate Plot	$col_{ctx}^+ + col_{agg} + res$		res
			Radar Chart	$col_{ctx}^+ + col_{agg} + res$		res
$[col_{ctx}^+, col_{met}, res]$	Correlation	$[N/T] * C * C$	Parallel Coordinate Plot	$col_{met} + col_{ctx}^+ + res$		res
			Radar Chart	$col_{met} + col_{ctx}^+ + res$		res

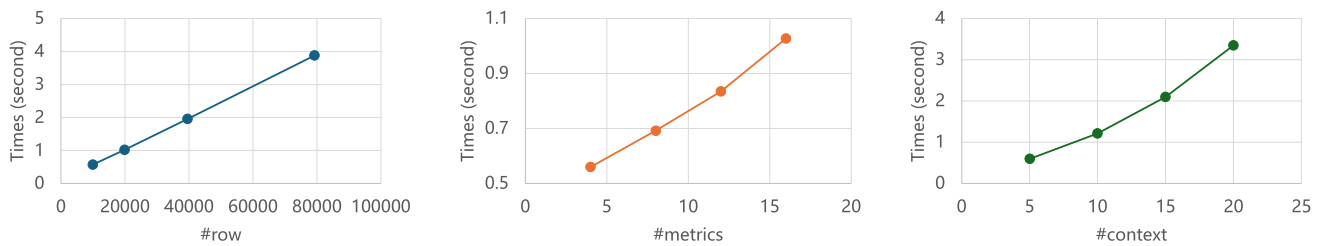


Fig. 3 Time Performance with fixed parameters (#metrics = 4, #context = 5 for varying rows; #context = 5, #rows = 9906 for varying metrics; #metrics = 4, #rows = 9906 for varying context). The figure

demonstrates that CAMV scales linearly with respect to both dataset size (number of rows) and the number of attributes (metrics and context), highlighting efficient performance across varying configurations

3.3 Prototype System

To demonstrate the feasibility and effectiveness of our framework, we developed a prototype system that enables analysts to explore and interpret analytical results through context-aware multi-view visualizations, as shown in Fig. 4. A high-resolution version is in the supplementary material.

The system allows users to upload both a result table and the original fact table, and to specify the associated analysis metadata—including aggregation attributes (col_{agg}), metrics (col_{met}), algorithm outputs (res), and the analysis algorithm (alg), as shown in (A). Once the data and metadata are provided, the system parses the input to construct analytical subspaces and the contextual features required for visualization. The interface consists of two coordinated panels: (B) displays visualizations of the analysis results, while (C) presents contextual visualizations derived from the augmented features.

To arrange the visualizations in a coherent and semantically meaningful layout, we compute the pairwise distances between views based on their visual encodings, underlying data, and the semantic similarity of their corresponding subspaces [57]. We then apply multi-view layout algorithms [26] and incorporate interaction coordination strategies [25] to support effective organization and exploration. Users can hover to highlight linked data elements, and click or use lasso to examine specific data points of interest—thereby facilitating the discovery of meaningful relationships between the analysis results and their contextual explanations.

4 Evaluation

In this section, we evaluate CAMV via time performance, a user study, a qualitative comparison, and a use case with existing multi-view recommendation systems.

4.1 Time Performance

To evaluate the efficiency of CAMV, we conducted performance experiments using real-world datasets in a tech company, varying three key parameters: the number of rows (10,000, 20,000, 40,000, and 80,000), the number of metrics for analysis (4, 8, 12, and 16), and the number of context attributes (5, 10, 15, and 20). For each configuration, we recorded the processing times to analyze scalability and algorithm complexity.

The results, shown in Fig. 3, demonstrate that CAMV scales linearly with respect to dataset size and attributes related to analysis. The average time complexity of the pipeline was measured as $O(n)$, where n represents the size of the dataset (number of rows or attributes for analysis). On average, processing time increased by approximately 1 second per additional 20,000 rows, or per 16 additional target attributes, or per 8 additional other attributes. For example, handling a dataset with 20,000 rows and 9 total attributes took around 1 second. These results demonstrate that CAMV scales linearly and efficiently, ensuring responsive interactions even with moderately large datasets common in real-world analysis scenarios.

4.2 Use Case

Bob, a data analyst, was exploring the *Missing Migrants* dataset from Kaggle¹, which comprises more than 13000 rows and 19 columns. This dataset comprises numerical, categorical, and temporal attributes, providing detailed information on migrant incidents, including demographic (such as *Number of Male* and *Number of Female*) and incident-related attributes (such as *Incident Type*, *Number of Dead*, and *Cause of Death*). Bob's primary goal was to identify the underlying risk factors that distinguish high-mortality migration regions from low-mortality ones, moving beyond simple statistical summaries to understand the “why” behind the clustering outcomes. He thus aggregated the data by *Region of Origin* and performed clustering

¹ <https://www.kaggle.com/code/sagorkumarmitra/global-missing-migrants-dataset/input>.

analysis based on key attributes, including *Number of Dead*, *Minimum Estimated Number of Missing*, *Total Number of Dead and Missing*, and *Number of Survivors*. The analysis categorized regions into three clusters labeled 0, 1, and 2.

To interpret these clustering results, Bob uploaded the original fact table and result table, and configured related analysis metadata into CAMV (shown in Fig. 4A). The system generated recommended visualizations, including result visualizations showing clustering outcomes (B) and contextual visualizations aiding interpretation (C). Due to the high dimensionality of aggregated target attributes, CAMV employed dimensionality reduction via principal component analysis (PCA), projecting the result table onto two principal components (*component 1* and *component 2*)

(Fig. 4B). Although three clusters were clearly separated in this reduced space, the abstract PCA components leave Bob unsure what these clusters mean and **why** these regions were grouped together.

Bob therefore examined the contextual visualizations provided by CAMV, including four stacked histograms (a-d), three grouped bar charts (e), and three scatter plots (f). From the grouped bar charts (e), Bob noted that regions in Cluster 0 had the highest numbers of dead and missing individuals but the lowest numbers of survivors. In contrast, regions in Cluster 1 had the lowest dead and missing rates and highest survival rates, with Cluster 2 positioned between them.



Fig. 4 The interface of CAMV. The user may upload data tables and specify columns of interest (A). Then, users may interact with the multiple-view visualization (C) to understand the contexts of the analysis result (B)

To uncover the underlying drivers, Bob analyzed the stacked histograms and observed that Cluster 0 exhibited notably high male migrant rates (a), incidents dispersed throughout all 12 months (b), higher longitudes (c), and diverse causes of death (d). These patterns suggest that migration routes originating from Cluster 0 regions are characterized by persistent, year-round risks across various conditions, where the high male participation likely reflects the demographic composition of migrants attempting these high-stakes, systemic-risk routes. Conversely, Cluster 1 showed highly concentrated incidents, mostly within a single month alongside more homogeneous causes of death, indicating specific underlying events. While most regions in Cluster 1 has lower male migrant rates, Bob noticed an outlier bar (a1) in histogram (a), representing a region in Cluster 1 with an unusually high male migrant rate. Upon clicking this specific bar, he discovered that it corresponded to Oceania (b1), which had only males but experienced a singular tragic event concentrated in one month (b2). This discovery allowed Bob to confirm that Cluster 1 was not merely defined by lower male participation, but by the temporal concentration of its incidents, distinguishing it from the persistent risks in Cluster 0. Bob then turned to the scatter plots (f) and validated the correlation between deaths and male participation rates. The scatter plots revealed a consistent pattern: regions with higher male participation had a positive correlation with higher death rates, confirming earlier observations from the bar charts (a).

Finally, Bob concluded that migration routes originating from Cluster 0 are characterized by persistent, year-round, and systemic dangers, where the high mortality is related to diverse causes, and scattered incidents throughout the year. In contrast, routes in Cluster 1 exhibit a sporadic risk profile, characterized by more concentrated incidents and homogeneous causes, suggesting that tragedy in these regions is driven by isolated events. Furthermore, he identified that

male migrant participation can serve as a demographic indicator reflecting these varying risk levels.

4.3 User Study

We conducted a within-subjects user study with 12 participants to assess the usability and effectiveness of CAMV, using ChatGPT (GPT-4o as the API) as the baseline system. Direct comparisons with existing multi-view systems are difficult due to differences in input formats and their lack of focus on result interpretation. Although GPT-4o cannot generate interactive dashboards, it was selected as the baseline given its widespread use in data analysis and ability to handle general inputs. Despite the inherent differences in output between GPT-4o and CAMV, a comparison study was conducted, as evaluating CAMV in isolation would limit the assessment. A more detailed qualitative comparison is provided in Sect. 4.4.

4.3.1 Participant

We recruited 12 participants via social media and word-of-mouth referrals (6 females, P1-12), with ages ranging from 22 to 33 years ($M = 26, SD = 3.10$). They were graduate students or senior engineers in a high-tech company. The average familiarity with data analysis was high ($M = 4.17, SD = 0.72$), and the familiarity with visualization was moderately high ($M = 4, SD = 0.74$). The detailed demographic information is shown in Table 5.

4.3.2 Study Materials

We selected the same *Missing Migrants* dataset used in the Use Case (Sect. 4.2). To prepare the data for the user study, we applied the KMeans clustering algorithm to categorize regions based on key metrics such as the number of deaths, number of males involved, and number of survivors. Additionally, we identified outliers using statistical methods (i.e., points beyond two standard deviations). These clustering and outlier detection results were provided to participants for interpretation during the study. We generated different analytical results from the same dataset to reduce participants' cognitive burden of familiarizing themselves with new data. For detailed analysis results, please refer to the supplementary materials.

4.3.3 Procedure

Each session lasted approximately 45 minutes. We began by introducing the study's purpose and procedure, followed by collecting participants' demographic data and familiarity with data analysis and visualizations. Participants then

Table 5 Participant demographics and self-reported familiarity scores

Participant	Gender	Age	Familiarity Data Analy.	Familiarity Data Vis.
P1	M	25	5	5
P2	F	26	4	4
P3	F	23	3	4
P4	M	23	4	4
P5	F	25	3	3
P6	M	22	5	3
P7	M	29	4	4
P8	M	26	4	4
P9	F	28	5	5
P10	M	33	4	3
P11	F	24	4	4
P12	F	28	5	5
Mean		26	4.17	4.00
SD		3.10	0.72	0.74

interacted with both the ChatGPT interface and CAMV, in a counterbalanced, random order to minimize learning effects. For CAMV, participants started with a 10-minute tutorial to familiarize themselves with the interface and functionalities, until they felt comfortable. Then they moved to the formal task, where they interpreted an analysis result using both GPT-4o and CAMV. Participants were encouraged to verbalize their thought processes using the think-aloud protocol. Upon completing the task, participants filled out a user experience questionnaire (Table 7) and participated in a semi-structured interview to provide qualitative feedback on both systems' usability and effectiveness.

4.3.4 Results

During the study, participants using CAMV required significantly less time to understand the analytical results compared to ChatGPT (355.0 vs. 887.9 seconds, $p < 0.005$). In the following sections, we present both quantitative and qualitative results of the user study.

Performance Metrics. Participants achieved a thorough understanding of the data analysis results after reviewing an average of 6.25 visualizations out of the 10 provided. We further measured the effectiveness of CAMV using the precision@K metric [58], assessing how many visualizations participants considered helpful among the top K recommended visualizations. Higher precision indicates more effective recommendations. As shown in Table 6, precision@K increased with larger K values, while normalized precision@K (precision@K/K) remained relatively stable at means of 0.94, 0.86, and 0.81 for K = 3, 6, and 10, respectively. This indicates that approximately 87% of the recommended visualizations were considered helpful for interpreting analysis results.

Questionnaire Results. Table 7 presents participants' ratings of both systems based on a 5-point Likert scale. For more details of the rating from participants, please refer to the folders in the supplementary. We performed a paired-T test to compare CAMV and the baseline in all aspects. Quantitative results show that CAMV significantly outperformed GPT-4o across all dimensions. Participants found CAMV to be more helpful in understanding analytical results, easier to use and comprehend, and more reliable than ChatGPT. They also reported higher confidence in their understanding of the results and expressed greater trust in the explanations provided by CAMV.

Qualitative Feedback. Participants found that both systems helped with understanding analytical results. However, they appreciated CAMV more for its non-linear, information-dense, and data-grounded reasoning. ChatGPT was praised for its intuitive natural language narration and external knowledge, but it was criticized for unverifiable claims,

Table 6 The table shows the Precision@K and normalized Precision@K (Precision@K/K) at K = 3, 6, and 10

Precision@K	AVG $\pm$ SD	Normalized AVG
P@3	2.83 $\pm$ 0.15	0.94
P@6	5.16 $\pm$ 0.52	0.86
P@10	8.08 $\pm$ 2.45	0.81
AVG		0.87

Precision@K increases with larger K values, while normalized Precision@K remained relatively stable. The consistently high normalized Precision@K indicates that approximately 87% of the recommended visualizations were considered effective for interpreting analysis results

Table 7 User Questionnaire Results (5-point Likert scale). CAMV significantly outperformed GPT-4o across all dimensions (with all $p < 0.01$)

Question	CAMV	ChatGPT	p-value
The system helps me understand the results	4.33 $\pm$ 0.40	3.17 $\pm$ 1.11	< 0.005
The system is easy to use	4.67 $\pm$ 0.28	3.83 $\pm$ 1.4	< 0.005
The system output is easy to understand	4.50 $\pm$ 0.2	3.75 $\pm$ 1.06	< 0.01
The system output is accurate	4.83 $\pm$ 0.39	3.83 $\pm$ 0.72	< 0.005
I trust the explanation provided by the system output	4.67 $\pm$ 0.49	3.83 $\pm$ 0.94	< 0.005
I am confident in my understanding of analytical results	4.17 $\pm$ 0.22	3.50 $\pm$ 0.80	< 0.005
I am willing to use the system in the future	4.50 $\pm$ 0.32	3.83 $\pm$ 1.40	< 0.005

a lack of direct data traceability, and verbose, unstructured responses. Thus, some participants (P1-2, P6-8, P12) expressed interest in integrating the advantage of GPT to CAMV, e.g., having more contextual cues or lightweight summaries alongside charts to enhance interpretability. P6 noted, "*ChatGPT provides external knowledge that help me grasp the potential reasons behind results. It would be great if CAMV could include this to improve interpretability.*"

Multi-view visualizations can offer a non-linear and information-dense presentation for better understanding. Participants found CAMV's MVs particularly valuable for offering a comprehensive overview of the data. In contrast, ChatGPT's lengthy narration and sparse visualizations were seen as less efficient in supporting understanding. P8 shared, "*MVs (in CAMV) provides a very information-dense context to understand the analysis results. I can see everything at once, rather than needing to get information linearly from GPT.*" P3 also noted, "*I need to prompt GPT to get the information step by step, then organize it manually, which is really ineffective and time-consuming.*"

Data-grounded reasoning can increase user trust. All participants expressed high levels of trust in CAMV due to its data-driven, transparent approach. P1 stated, "*I really trust the results from CAMV because everything is data-grounded*". Conversely, ChatGPT was criticized for

generating responses that lacked data traceability, leading to reduced trust. P6 explained, “*I like ChatGPT’s explanation, but I can’t fully trust it. I need to spend lots of time validating what the findings come from or whether the data in the recommended visualization is true. This is really tedious especially when the data is complex.*”

Interactive visualizations can facilitate deeper data exploration and understanding. Participants greatly appreciated the interactivity of CAMV, especially in comparison to GPT, which lacked interactive capabilities and required participants to manually piece together the information. P7 stated, “*I feel more engaged with CAMV because I can interact with the visualizations and drill deeper into the data.*” P12 added, “*With GPT, I just receive static answers without being able to interact with the data. This makes it harder to understand relationships between the GPT answers.*”

4.4 Qualitative Comparison

CAMV differs fundamentally from existing multi-view visualization recommendation systems in both input format and design goals. While prior systems typically take raw datasets and optionally user-specified tasks or attributes as input [12, 14, 15], CAMV instead accepts users’ analysis results and associated metadata, aiming to support result interpretation rather than general-purpose exploration. As no existing systems are designed to take analytical results as input and recommend multi-view visualizations tailored for explanation, it would be inappropriate and unfair to treat these systems as baselines. Therefore, we provide a qualitative comparison to highlight differences through both a theoretical and a scenario-based comparison.

Theoretical Comparison. Regarding **intent specification**, prior systems such as Medley [14] or TaskVis [15] require users to manually specify analysis tasks and attributes of interest. In contrast, CAMV automatically infers the implicit intent behind analytical results, freeing users from the need for prior knowledge about what is required to interpret those results. In terms of the **workflow of multi-view recommendation**, data-driven methods like MultiVision [12] optimize multi-view composition by maximizing data coverage or insight diversity from the raw dataset without considering users’ current analytical focus. CAMV, however, starts from specific analysis outputs and explicitly recommends contextual subspaces that help explain them.

Moreover, **interfaces powered by large language models (LLMs)**, such as ChatGPT, represent another paradigm by accepting free-form user intent in natural language along with multimodal data (e.g., tables and charts or images). However, their **responses often consist of a loosely connected collection of visualizations** lacking coordination

and interactivity, which are key features of multi-view systems. This absence of structural coherence limits users’ ability to synthesize insights and follow up with focused exploration.

Scenario-based Comparison. To further illustrate the practical advantage of CAMV, we consider the result interpretation of the *Missing Migrants* dataset described in the Use Case in Section 4.2.

CAMV vs. MV Recommendation Systems. A data-driven system like MultiVision [12] or DashBot [13] would prioritize subspaces with high statistical significance or diversity across the entire dataset. For instance, it might recommend a scatter plot showing the strong correlation between *Total Number of Dead and Missing* and *Number of Dead*. While statistically interesting, these views are **result-agnostic** and do not explain why specific regions were grouped into Cluster 0 versus Cluster 1. Furthermore, intent-driven methods like TaskVis [15] or Medley [14] require users to manually specify relevant attributes, which is difficult when the user is still in the dark about which features (e.g., gender or temporal patterns) drive the cluster characteristics. In contrast, CAMV treats the analytical result as user intent, identifying that the diversity of death causes and temporal distribution of incidents are key to distinguishing these clusters. By surfacing this “targeted evidence”, CAMV provides the necessary context to interpret clustering outcomes.

CAMV vs. LLM-powered Interface. In our user study, when users prompted GPT-4o to explain the clustering results by providing both the result table and original data tables, the model frequently generated charts and descriptions that merely summarized the result table (e.g., noting that “Cluster 0 has high mortality rates”). It often failed to proactively explore the original data context to find the reasons behind the clusters. Consequently, users had to engage in a tedious, iterative dialogue to refine their prompts. This process becomes particularly challenging when users struggle to interpret the LLM outputs and then articulate the precise queries needed to guide the LLM. Conversely, CAMV provides a comprehensive suite of visualizations that automatically incorporate the original data as evidence. By presenting all relevant views at once, CAMV enables non-linear exploration, removing the need for users to manually refine their intent through multiple conversational turns. Furthermore, as the LLM-generated visualizations were static and independent, users struggled to link specific data points and synthesize insights across different charts. For example, in our case study, when Bob encountered an unexpectedly high male rate in Cluster 1, an LLM-based workflow may require him to manually bridge his prior understanding with this new anomaly through complex prompt engineering. In contrast, CAMV leverages coordinated interactivity to facilitate a seamless transition from anomaly detection to hypothesis

verification. By directly clicking a specific visual element, Bob instantly identified that the anomaly represented *Oceania*, confirming that this anomaly still aligns with the cluster characterized by highly concentrated incidents happened in a single month and homogeneous causes of death. The coordinated MVs are far more efficient than through isolated LLM responses.

Overall, CAMV provides a streamlined experience centered on understanding analytical results. It focuses on the MVs and enables more targeted visual explanations and reduces the user's cognitive burden of specifying what is needed for result interpretation, aligning more closely with practical analytical scenarios.

5 Discussion

5.1 Design Implications

Rethinking evaluation metrics for multi-view visualization recommendation. We currently adopt standard metrics (precision@K) for recommender systems to assess the effectiveness of subspace recommendation. However, this approach assumes a linear and independent consumption model, which does not align with how users interact with MVs. In practice, visualizations are spatially organized, semantically linked, and interpreted as a whole, not as isolated recommendations. To meaningfully assess the value of multi-view recommendations, future work should move beyond rank-based evaluation and toward metrics that capture perceptual coherence, narrative flow, and task relevance, ultimately aligning evaluation with the goal of multi-perspective insight construction.

Redefining what counts as important insights. CAMV prioritizes contextual subspaces with high divergence across groups-patterns typically regarded as interesting [59]. However, our user study revealed that participants also valued subspaces with low variance across groups, particularly for validating assumptions. For example, in a mortality analysis case, participants found that the number of death causes varied across categories and wished to further examine the specific causes of death within each group. Although the system had omitted this contextual subspace due to low divergence, participants viewed it as essential for confirming that cause-of-death distributions were indeed irrelevant. This highlights that multi-view visualization should support not only the discovery of surprising patterns, but also the verification of invariants. Future research should go beyond traditional interestingness metrics and treat neutral, low-variance patterns as equally valuable.

5.2 Generalizability

Task and data applicability. CAMV currently supports analysis tasks with discrete output labels, such as clustering, classification (binary and multi-class), outlier detection, and discretization-based binning, as CAMV identifies important attributes through the distributional divergence across result groups. The system is compatible with general tabular datasets containing numerical, categorical, or temporal attributes, as our feature enhancement module (Sect. 3.1.2) is specifically designed to handle these data types. Currently, CAMV does not support tasks with continuous outcomes (e.g., regression). The continuous outcomes pose a significant challenge for contrastive explanation, as small numerical fluctuations are often harder to interpret than distinct group boundaries. Future work may explore collaboration with domain experts to discretize continuous outcomes into interpretable intervals, or investigate alternative explanation techniques to identify meaningful data contexts for continuous results.

Visualization rule coverage and mapping. Our current implementation includes five representative chart types covering common subspaces for result interpretation. The rule tables (Table 4a and b) are modular and can be easily extended to other data subspaces beyond the current implementation (e.g., histograms for distributions, scatterplots for correlations). We currently support two types of user intent due to limited empirical grounding of other intents. Future work may explore a broader range of realistic intents to enhance our framework.

Moreover, our current visualization method relies on predefined heuristic rules for visual mapping. While this ensures predictable and effective encodings for standard data facts, it may lack the flexibility to adapt to unique domain requirements or unconventional data distributions. Future work could explore integrating data-driven recommendation models to offer more adaptive and personalized visual explanations.

Downstream applications. CAMV is applicable when an analytics pipeline produces discrete outputs (e.g., clustering, classification, and outlier detection), and analysts need an evidence-grounded understanding of the results. Accordingly, CAMV can be integrated into downstream applications where discrete labels or groups require additional contextualization for interpretation, such as stratified user cohorts in recommender systems [60, 61], resilience monitoring in transportation [62], alert categories in healthcare [63], community assignments in knowledge-graph workflows [64], and error diagnosis in multi-agent systems [65].

5.3 Limitations and Future Work

While CAMV provides effective multi-view visualizations for result interpretation, several areas remain open for enhancement. First, although parameters like top- M important features and equal scoring weights were empirically set to manage cognitive load, their sensitivity has not been formally tested. This is because CAMV produces interpretation-oriented recommendations without an established ground truth, and thus evaluating changes in interpretability and usefulness typically requires substantial human studies. Future research should evaluate how varying these parameters affect the ranking of data subspaces and the overall interpretability of the results, and explore scalable evaluation protocols and quantitative proxies to support such analysis. Second, CAMV currently lacks advanced interactions such as drill-down across coordinated views, which could support deeper and more flexible analysis [31]. Third, onboarding mechanisms such as guided tours or progressive disclosure [66] may ease the learning curve for first-time users navigating multi-view layouts. Finally, integrating CAMV with large language models—leveraging their external knowledge and ability to generate natural language narratives—may further enhance user engagement and interpretability [33].

6 Conclusion

We presented CAMV, a context-aware multi-view visualization framework designed to support the interpretation of analytical results by automatically inferring users' analytical intents and recommending relevant visualizations. By treating analysts' result tables and associated metadata as implicit intents, CAMV identifies explanatory subspaces that facilitate result interpretation and presents them through coordinated multi-view visualizations. The prototype system enables users to interactively explore these contextual subspaces, and our user study demonstrates that CAMV effectively provides rich contextual information and thus facilitates analytical understanding. We hope our work emphasizes the importance of enabling efficient multi-perspective and context-aware information access for interpreting analytical results.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s41019-026-00353-x>.

Acknowledgements This work is partially supported by Grant No. 22230930Q007

Author contributions Yanna Lin proposed the core idea, designed the methodology and evaluation, led the writing, and created the fig-

ures. Liwenhan Xie contributed to the implementation of core methods, interface design, and figurecreation. Leixian Shen conducted the evaluation study and contributed to parts of the writing. Zhuochen Jin consulted with data analysis experts to provide methodological insights and helped review and refine the methodology. Sicheng Song contributed to the implementation of parts of the method and supported manuscript writing. Zikun Deng contributed to the discussion section and polished the writing. Ke Xu provided funding support, advised on the conceptual development, and helped polish the manuscript. All authors read and approved the final manuscript.

Funding This research project is supported by TC20221227086.

Data Availability The data generated during the user study and the demonstration materials will be made publicly available. However, since this project is funded by a company, the system code cannot be released upon publication.

Declarations

Competing Interests The authors declare that they have no competing interests

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Roberts JC (2007) State of the art: Coordinated & multiple views in exploratory visualization. In: Proceedings of the Fifth International Conference on Coordinated and Multiple Views in Exploratory Visualization, pp. 61–71. IEEE, Piscataway, NJ, USA. <https://doi.org/10.1109/CMV.2007.20>
2. Cheng F, Liu D, Du F, Lin Y, Zytek A, Li H, Qu H, Veeramachaneni K (2021) Vbridge: connecting the dots between features and data to explain healthcare models. *IEEE Trans Vis Comput Graph* 28(1):378–388
3. Cao N, Lin C, Zhu Q, Lin Y-R, Teng X, Wen X (2017) Voila: visual anomaly detection and monitoring with streaming spatiotemporal data. *IEEE Trans Vis Comput Graph* 24(1):23–33. <https://doi.org/10.1109/TVCG.2017.2744419>
4. Eichmann P, Solleza F, Tatbul N, Zdonik S (2019) Visual exploration of time series anomalies with metro-viz. In: Proceedings of the 2019 International Conference on Management of Data, pp. 1901–1904. ACM, New York, NY, USA. <https://doi.org/10.1145/3299869.3320247>
5. Wang J, Suzuki A, Xu L, Tian F, Yang L, Yamanishi K (2019) Orderly subspace clustering. *Proceedings of the AAAI Conference on Artificial Intelligence* 33:5264–5272

6. Lundberg SM, Lee S-I (2017) A unified approach to interpreting model predictions. *Advances in neural information processing systems* 30
7. Chen X, Zeng W, Lin Y, Ai-Manee HM, Roberts J, Chang R (2021) Composition and configuration patterns in multiple-view visualizations. *IEEE Trans Vis Comput Graph* 27(2):1514–1524. <https://doi.org/10.1109/TVCG.2020.3030338>
8. Liu W, Chen Y, Yue X, Zhang C, Xie S (2023) Safe multi-view deep classification. *Proceedings of the AAAI Conference on Artificial Intelligence* 37:8870–8878
9. Wan Z, Zhang C, Zhu P, Hu Q (2021) Multi-view information-bottleneck representation learning. *Proceedings of the AAAI Conference on Artificial Intelligence* 35:10085–10092
10. Satyanarayan A, Moritz D, Wongsuphasawat K, Heer J (2016) Vega-Lite: a grammar of interactive graphics. *IEEE Trans Vis Comput Graph* 23(1):341–350
11. Hunter JD (2007) Matplotlib Development Team: Matplotlib: Python plotting. Accessed: 2026-01-05. <https://matplotlib.org/>
12. Wu A, Wang Y, Zhou M, He X, Zhang H, Qu H, Zhang D (2022) MultiVision: designing analytical dashboards with deep learning based recommendation. *IEEE Trans Vis Comput Graph* 28(1):162–172. <https://doi.org/10.1109/TVCG.2021.3114826>
13. Deng D, Wu A, Qu H, Wu Y (2023) DashBot: insight-driven dashboard generation based on deep reinforcement learning. *IEEE Trans Vis Comput Graph* 29(1):690–700. <https://doi.org/10.1109/TVCG.2022.3209468>
14. Pandey A, Srinivasan A, Setlur V (2023) MEDLEY: intent-based recommendations to support dashboard composition. *IEEE Trans Vis Comput Graph* 29(1):1135–1145. <https://doi.org/10.1109/TVCG.2022.3209421>
15. Shen L, Shen E, Tai Z, Xu Y, Dong J, Wang J (2022) Visual data analysis with task-based recommendations. *Data Sci Eng* 7(4):354–369
16. Zhao Y, Zhang Y, Zhang Y, Zhao X, Wang J, Shao Z, Turkay C, Chen S (2024) Leva: using large language models to enhance visual analytics. *IEEE Trans Vis Comput Graph* 31(3):1830–1847
17. Zhao Y, Wang J, Xiang L, Zhang X, Guo Z, Turkay C, Zhang Y, Chen S (2024) Lightva: lightweight visual analytics with llm agent-based task planning and execution. *IEEE Trans Vis Comput Graph* 31(9):6162–6177
18. Zhao Y, Shu X, Fan L, Gao L, Zhang Y, Chen S (2025) Proactiveva: Proactive visual analytics with llm-based ui agent. *arXiv preprint arXiv:2507.18165*
19. Bach B, Freeman E, Abdul-Rahman A, Turkay C, Khan S, Fan Y, Chen M (2023) Dashboard design patterns. *IEEE Trans Vis Comput Graph* 29(1):342–352. <https://doi.org/10.1109/TVCG.2022.3209448>
20. Sarikaya A, Correll M, Bartram L, Tory M, Fisher D (2019) What do we talk about when we talk about dashboards? *IEEE Trans Vis Comput Graph* 25(1):682–692. <https://doi.org/10.1109/TVCG.2018.2864903>
21. Qu Z, Hullman J (2018) Keeping multiple views consistent: constraints, validations, and exceptions in visualization authoring. *IEEE Trans Vis Comput Graph* 24(1):468–477. <https://doi.org/10.1109/TVCG.2017.2744198>
22. Setlur V, Correll M, Satyanarayan A, Tory M (2024) Heuristics for supporting cooperative dashboard design. *IEEE Trans Vis Comput Graph* 30(1):370–380. <https://doi.org/10.1109/TVCG.2023.3327158>
23. Tory M, Bartram L, Fiore-Gartland B, Crisan A (2023) Finding their data voice: practices and challenges of dashboard users. *IEEE Comput Graph Appl* 43(1):22–36. <https://doi.org/10.1109/MCG.2021.3136545>
24. Ma R, Mei H, Guan H, Huang W, Zhang F, Xin C, Dai W, Wen X, Chen W (2021) LADV: deep learning assisted authoring of dashboard visualizations from images and sketches. *IEEE Trans Vis Comput Graph* 27(9):3717–3732. <https://doi.org/10.1109/TVCG.2020.2980227>
25. Lin Y, Li H, Wu A, Wang Y, Qu H (2024) DMiner: dashboard design mining and recommendation. *IEEE Trans Vis Comput Graph* 30(7):4108–4121. <https://doi.org/10.1109/TVCG.2023.3251344>
26. Zhao J, Xu S, Chandrasegaran S, Bryan C, Du F, Mishra A, Qian X, Li Y, Ma KL (2023) Chartstory: automated partitioning, layout, and captioning of charts into comic-style narratives. *IEEE Trans Vis Comput Graph* 29(2):1384–1399. <https://doi.org/10.1109/TVCG.2021.3114211>
27. Chen J, Kong L, Wei H, Liu C, Ge Z, Zhao L, Sun J, Han C, Zhang X (2024) OneChart: Purify the Chart Structural Extraction via One Auxiliary Token. In: *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 147–155. ACM, New York, NY, USA. <https://doi.org/10.1145/3664647.3681167>
28. Li D, Yin H, Wang C, Song S, Li K, Li C (2021) Opinionmanager: Visual exploration of online reviews in p2p accommodation. In: *Proceedings of the 14th International Symposium on Visual Information Communication and Interaction*, pp. 1–8
29. Li D, Yin H, Wang C, Song S, Li K, Li C (2022) Visual recommendation for peer-to-peer accommodation with online reviews based on sentiment analysis and topic models. *J Vis* 25(6):1309–1327
30. Li H, Ying L, Zhang H, Wu Y, Qu H, Wang Y (2023) Notable: On-the-fly Assistant for Data Storytelling in Computational Notebooks. In: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI'23, pp. 1–16. ACM, New York, NY, USA. <https://doi.org/10.1145/3544548.3580965>
31. Cao Y-R, Li X-H, Pan J-Y, Lin W-C (2022) Visguide: User-oriented recommendations for data event extraction. In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA. <https://doi.org/10.1145/3491102.3517648>
32. Shen L, Shen E, Luo Y, Yang X, Hu X, Zhang X, Tai Z, Wang J (2023) Towards natural language interfaces for data visualization: a survey. *IEEE Trans Vis Comput Graph* 29(6):3121–3144. <https://doi.org/10.1109/TVCG.2022.3148007>
33. Lin Y, Li H, Yang L, Wu A, Qu H (2024) Inksight: leveraging sketch interaction for documenting chart findings in computational notebooks. *IEEE Trans Vis Comput Graph* 30(1):944–954. <https://doi.org/10.1109/TVCG.2023.3327170>
34. Song S, Zhang Y, Lin Y, Qu H, Wang C, Li C (2025) Gvvst: image-driven style extraction from graph visualizations for visual style transfer. *IEEE Trans Vis Comput Graph* 31(9):5975–5989. <https://doi.org/10.1109/TVCG.2024.3485701>
35. Yu Y, Shen L, Long F, Qu H, Chen H (2024) PyGWalker: On-the-fly Assistant for Exploratory Visual Data Analysis. In: *Proceedings of IEEE Visualization and Visual Analytics*, pp. 6–10. IEEE, Piscataway, NJ, USA. <https://doi.org/10.1109/VIS55277.2024.00009>
36. Xie L, Zhou Z, Yu K, Wang Y, Qu H, Chen S (2023) Wakey-wakey: Animate text by mimicking characters in a gif. In: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, New York, NY, USA. <https://doi.org/10.1145/3586183.3606813>
37. Shen L, Shen E, Tai Z, Wang Y, Luo Y, Wang J (2022) GALVIS: Visualization Construction through Example-Powered Declarative Programming. In: *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, pp. 4975–4979. ACM, New York, NY, USA. <https://doi.org/10.1145/3511808.3557159>
38. Epperson W, Jung-Lin Lee D, Wang L, Agarwal K, Parameswaran AG, Moritz D, Perer A (2022) Leveraging analysis history for improved in situ visualization recommendation. *Comput Graph Forum* 41(3):145–155. <https://doi.org/10.1111/cgf.14529>

39. Lalle S, Toker D, Conati C (2021) Gaze-driven adaptive interventions for magazine-style narrative visualizations. *IEEE Trans Vis Comput Graph* 27(6):2941–2952. <https://doi.org/10.1109/TVCG.2019.2958540>
40. Chen Z, Song S, Shum K, Lin Y, Sheng R, Qu H (2025) Unmasking deceptive visuals: Benchmarking multimodal large language models on misleading chart question answering. *EMNLP*, 1–31 <https://doi.org/10.48550/arXiv.2503.18172>
41. Song S, Zhang Y, Chen Z, Qu H, Wang C, Li C (2026) Vizdefender: Unmasking visualization tampering through proactive localization and intent inference, 1–11
42. Kho AN, Hayes MG, Rasmussen-Torvik L, Pacheco JA, Thompson WK, Armstrong LL, Denny JC, Peissig PL, Miller AW, Wei W-Q (2012) Use of diverse electronic medical record systems to identify genetic risk for type 2 diabetes within a genome-wide association study. *J Am Med Inform Assoc* 19(2):212–218
43. Payrovnaziri SN, Chen Z, Rengifo-Moreno P, Miller T, Bian J, Chen JH, Liu X, He Z (2020) Explainable artificial intelligence models using real-world electronic health record data: a systematic scoping review. *J Am Med Inform Assoc* 27(7):1173–1185
44. Ming Y, Qu H, Bertini E (2018) Rulematrix: visualizing and understanding classifiers with rules. *IEEE Trans Vis Comput Graph* 25(1):342–352. <https://doi.org/10.1109/TVCG.2018.2864812>
45. Che Z, Purushotham S, Khemani R, Liu Y (2017) Interpretable deep models for icu outcome prediction. In: *AMIA Annual Symposium Proceedings*, vol. 2016, p. 371
46. Ribeiro MT, Singh S, Guestrin C (2016) “why should i trust you?” explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144
47. Xu K, Yuan J, Wang Y, Silva C, Bertini E (2021) mtseer: Interactive visual exploration of models on multivariate time-series forecast. In: *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA. <https://doi.org/10.1145/3411764.3445083>
48. Lee DJ-L, Tang D, Agarwal K, Boonmark T, Chen C, Kang J, Mukhopadhyay U, Song J, Yong M, Hearst MA, Parameswaran AG (2021) Lux: always-on visualization recommendations for exploratory dataframe workflows. *Proc VLDB Endow* 15(3):727–738. <https://doi.org/10.14778/3494124.3494151>
49. Ma P, Ding R, Wang S, Han S, Zhang D (2023) InsightPilot: An LLM-empowered automated data exploration system. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 346–352. ACL, Singapore. <https://doi.org/10.18653/v1/2023.emnlp-demo.31>
50. Hubert L, Arabie P (1985) Comparing partitions. *J Classif* 2(1):193–218
51. Jaccard P (1901) Étude comparative de la distribution florale dans une portion des alpes et des jura. *Bull Soc Vaud Sci Nat* 37:547–579
52. Rousseeuw PJ (1987) Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *J Comput Appl Math* 20:53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
53. Heath TL (1956) *The Thirteen Books of Euclid's Elements*. Dover Publications, New York
54. Mackinlay J (1986) Automating the design of graphical presentations of relational information. *ACM Trans Graph* 5(2):110–141. <https://doi.org/10.1145/22949.22950>
55. Mackinlay J, Hanrahan P, Stolte C (2007) Show me: automatic presentation for visual analysis. *IEEE Trans Vis Comput Graph* 13(6):1137–1144. <https://doi.org/10.1109/TVCG.2007.70594>
56. Wongsuphasawat K, Moritz D, Anand A, Mackinlay J, Howe B, Heer J (2015) Voyager: exploratory analysis via faceted browsing of visualization recommendations. *IEEE Trans Vis Comput Graph* 22(1):649–658. <https://doi.org/10.1109/TVCG.2015.2467191>
57. Kim Y, Wongsuphasawat K, Hullman J, Heer J (2017) Graphscape: A model for automated reasoning about visualization similarity and sequencing. In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, pp. 2628–2638. ACM, New York, NY, USA. <https://doi.org/10.1145/3025453.3025866>
58. Järvelin K, Kekäläinen J (2002) Cumulated gain-based evaluation of ir techniques. *ACM Transactions on Information Systems (TOIS)* 20(4):422–446. <https://doi.org/10.1145/582415.582418>
59. Wang Y, Sun Z, Zhang H, Cui W, Xu K, Ma X, Zhang D (2019) Datasheet: automatic generation of fact sheets from tabular data. *IEEE Trans Vis Comput Graph* 26(1):895–905. <https://doi.org/10.1109/TVCG.2019.2934398>
60. Song J, Li Y, Li D, Wang X, Zhang R, Zhang H, Chen J (2026) Integrating deep clustering and multi-view graph neural networks for recommender system. *Knowl Based Syst*. <https://doi.org/10.1016/j.knosys.2026.115449>
61. Li D, Lu J, Wang Z, Wang J, Wang X, Shi F, Liu Y (2025) Recommender system based on noise enhancement and multi-view graph contrastive learning. *Appl Soft Comput* 177:113220
62. Zhang H, Ouyang M, Xu M, Sun W, Hong L, Li D (2026) Travel time resilience of national multimodal transport systems under extreme events: a passenger-oriented framework. *Reliab Eng Syst Saf*. <https://doi.org/10.1016/j.ress.2026.112256>
63. Sheng R, Wang Y, Wang X, Dai S, Guo Q, Peng T-Q, Qu H, Liu D (2025) Eminds: Understanding user behavior progression for mental health exploration on social media. *IEEE Transactions on Visualization and Computer Graphics*
64. Wang J, Li D, Du X, Li H, Hu Z (2026) Knowledge-based visual question classification using quaternion hypergraph consistent network. *Inf Process Manag* 63(4):104591
65. Sheng R, Yang Y, Shi C, Lin Y, Chen Z, Qu H, Cheng F (2026) Dills: Interactive diagnosis of llm-based multi-agent systems via layered summary of agent behaviors. *arXiv preprint arXiv:2602.05446*
66. Dhanoa V, Walchshofer C, Hinterreiter A, Stitz H, Groeller E, Streit M (2022) A process model for dashboard onboarding. *Comput Graph Forum* 41(3):501–513. <https://doi.org/10.1111/cgf.14558>